

A user-defined function called “checkDroneArm” was written and used to analyze different designs. The results were checked with SolidWorks static simulation.

The function does both the thrust-to-weight and the FE analysis. The thrust-to-weight analysis gives the result of the max payload, which the challenge dictates should be above 0.5 kg. The function then uses MatLabs built in FEA.

SolidWorks simulation is used as a sanity check to verify the data from the MatLab FEA. The material compared is aluminum alloy.

Results for the Flat Arm Design:

name	mass	cost	max_payload	max_stress	SF	max_displacement
"Carbon Fiber Composite (CFRP)"	0.065726	1.1091	0.7371	2.0921e+07	28.68	0.00058518
"Aluminum Alloy"	0.11091	0.01733	0.55635	2.07e+07	13.285	0.00059239
"Fiberglass Composite (GFRP)"	0.07805	0.13864	0.6878	2.1072e+07	14.237	0.0016406
"PLA Plastic"	0.051348	0.0017612	0.79461	2.0493e+07	2.9278	0.011651
"ABS Plastic"	0.043133	0.0017612	0.82747	2.056e+07	1.9455	0.020406
"Wood (Birch)"	0.024647	0.00032863	0.90141	2.056e+07	3.8911	0.0040811

Results for the Triangle Arm Design:

name	mass	cost	max_payload	max_stress	SF	max_displacement
"Carbon Fiber Composite (CFRP)"	0.053967	0.91069	0.78413	2.464e+07	24.351	0.00088486
"Aluminum Alloy"	0.091069	0.01423	0.63572	2.4737e+07	11.117	0.00089779
"Fiberglass Composite (GFRP)"	0.064086	0.11384	0.74366	2.4577e+07	12.206	0.0024772
"PLA Plastic"	0.042162	0.0014461	0.83135	2.4839e+07	2.4156	0.017698
"ABS Plastic"	0.035416	0.0014461	0.85834	2.4804e+07	1.6126	0.030973
"Wood (Birch)"	0.020238	0.00026983	0.91905	2.4804e+07	3.2252	0.0061947

Max Stress Error: 0.774%

Max Displacement Error: 0.021%

Results for the Modified Triangle Arm Design:

name	mass	cost	max_payload	max_stress	SF	max_displacement
"Carbon Fiber Composite (CFRP)"	0.043238	0.72965	0.82705	2.5063e+07	23.94	0.00090154
"Aluminum Alloy"	0.072965	0.011401	0.70814	2.516e+07	10.93	0.00091497
"Fiberglass Composite (GFRP)"	0.051346	0.091206	0.79462	2.5001e+07	11.999	0.0025235
"PLA Plastic"	0.03378	0.0011587	0.86488	2.5262e+07	2.3751	0.018042
"ABS Plastic"	0.028375	0.0011587	0.8865	2.5227e+07	1.5856	0.031572
"Wood (Birch)"	0.016214	0.00021619	0.93514	2.5227e+07	3.1712	0.0063144

Max Stress Error: 1.53%

Max Displacement Error: 0.102%

Results for beam arm V1:

name	mass	cost	max_payload	max_stress	SF	max_displacement
"Carbon Fiber Composite (CFRP)"	0.044239	0.74654	0.82304	1.3397e+07	44.786	0.00021593
"Aluminum Alloy"	0.074654	0.011665	0.70138	1.3441e+07	20.46	0.00021864
"Fiberglass Composite (GFRP)"	0.052534	0.093317	0.78986	1.3362e+07	22.451	0.0006053
"PLA Plastic"	0.034562	0.0011855	0.86175	1.3463e+07	4.4566	0.0043007
"ABS Plastic"	0.029032	0.0011855	0.88387	1.3459e+07	2.9719	0.0075321
"Wood (Birch)"	0.01659	0.0002212	0.93364	1.3459e+07	5.9439	0.0015064

Max Stress Error: 3.05%

Max Displacement Error: 0.3%

Results for Circle Arm V1

name	mass	cost	max_payload	max_stress	SF	max_displacement
"Carbon Fiber Composite (CFRP)"	0.04435	0.74841	0.8226	5.061e+06	118.55	3.4907e-05
"Aluminum Alloy"	0.074841	0.011694	0.70064	4.887e+06	56.271	3.4783e-05
"Fiberglass Composite (GFRP)"	0.052666	0.093551	0.78934	5.0864e+06	58.981	9.7355e-05
"PLA Plastic"	0.034649	0.0011884	0.86141	4.9392e+06	12.148	0.0007001
"ABS Plastic"	0.029105	0.0011884	0.88358	4.9846e+06	8.0247	0.0012295
"Wood (Birch)"	0.016631	0.00022175	0.93347	5.0252e+06	15.92	0.00024756

Max Stress Error: 12.3%

Max Displacement Error: 0.287%